DBMS Notes

Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It works by dividing large tables into smaller related tables and defining relationships between them. The goal is to avoid insertion, update and deletion anomalies.

Anomalies

An insertion anomaly happens when we cannot add data without adding unrelated data. An update anomaly happens when the same fact is stored in many rows and changing it in one row leaves the others inconsistent. A deletion anomaly happens when deleting one fact accidentally removes another useful fact.

First Normal Form (1NF)

A table is in 1NF if every column holds atomic (indivisible) values and there are no repeating groups. For example, a column storing 'Math, Physics' for a student violates 1NF. Fix it by creating one row per subject.

Second Normal Form (2NF)

A table is in 2NF if it is in 1NF and every non-key attribute depends on the whole primary key, not just part of it. Partial dependency only occurs with composite keys. To fix it, move the partially dependent attributes into a separate table.

Third Normal Form (3NF)

A table is in 3NF if it is in 2NF and no non-key attribute depends on another non-key attribute. This removes transitive dependency. Example: if StudentID -> DeptID and DeptID -> DeptName, then DeptName should move to a Department table.

BCNF

Boyce-Codd Normal Form is a stricter version of 3NF. A table is in BCNF if for every functional dependency X -> Y, X is a super key. BCNF removes the remaining anomalies caused by overlapping candidate keys.

Transactions and ACID

A transaction is a unit of work that must be completed fully or not at all. ACID stands for Atomicity, Consistency, Isolation and Durability. Atomicity means all or nothing. Consistency means the database moves from one valid state to another. Isolation means concurrent transactions do not interfere. Durability means committed data survives crashes.